LARGE COMMERCIAL SODIUM COOLED FAST REACTORS

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1. Introduction

The aims of this paper are to identify and discuss some of the principas technical and economic features that may be expected of large commercial fast reactors, and to show how these requirements will be met by designe developed from—and substantiated by—the prototype described in some detail in Paper 3/3. The contents of the paper are largely drawn from a 3-year study of commercial units based on the prototype concept, undertaken jointly by the three industrial organizations with which the authors are associated.

The study, conducted at the invitation of the U.K. Atomic Energy Authority and completed last summer, forms part of a continuing programme of close collaboration between the Authority and British industry, on design and development for the fast breeder project. It was timed to coincide with the formative stage of the prototype design and aimed both at confirming the economic incentive for the system and at ensuring that the prototype embraced, as far as possible, all the important features which were likely to appear in subsequent commercial stations. To help meet these objectives, the design of a 2 × 1000 MW(e) station was worked up in detail, and costed. An outline description of this design, together with a list of the major parameters, has been included in section 4. Since the major components are generally similar to those of the prototype, a detailed description has not been included.

Naturally, it is not expected that present thinking can embrace all the trends in the development of fast breeders over the next 10 years. Many changes in detail, and some, perhaps, in principle, will occur. Unless unsuspected technical or safety problems arise, these changes will always tend

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to improve, or at least maintain the economic position, and so a detailed design prepared at this stage forms a useful basis for comparison.

2. Choice of Some Major Features

In shaping the design and assessing the prospects of any power reactor system, the major considerations are, of course, how low the capital and generation costs can be made and how they will compare with those of known competitors. Clearly, all technical features influence the answers to these questions, and before discussing the overall costs it is worth while reviewing a few of the more important of these features, from the points of view of what is desired and what can be achieved.

Fuel and breeder elements for the commercial stations will be developed by UKAEA, and proved in the prototype. The design and performance of these items has been fully covered in other papers and it is not proposed to discuss them further here, except in relation to some general remarks on breeding and steam temperatures.

The requirements have to be considered in relation to the time scale. The prototype is scheduled for completion in 1970, and full load operation in mid-1971. Allowing perhaps 2 years for initial demonstration of the fuel and novel components, inquiries for the first commercial station may be expected by 1973. With 4 years' construction period, this station could be commissioned in 1978.

Unit Size

Current trends in the United Kingdom suggest that by this time the interconnected grid will be large enough to accept units in the range 1000 MW(e)–1500 MW(e) and, by the mid-1980's, 2000 MW(e). In so far as an increase in size brings economies, then provided reliability is no problem, there will be a demand for units of this capacity. Indications are that similar general trends towards larger units exist in continental Europe[1] and in the United States.[2] The actual maximum sizes of the reactors built will depend to some extent on the state of turbine development at the time, as each reactor will supply an integral number. For smaller installations, the minimum unit sizes will be set by the point at which, with the higher capital cost component characteristic of nuclear, as opposed to fossil, fuelled plants, the design ceases to be competitive.

Recognizing that the prototype itself would form a useful basis from which to assess the commercial prospects of smaller units, the desing studies for commercial units were concentrated at the larger end of the scale, with a 2×1000 MW(e) station as a reference. Starting today, it is probable that a somewhat larger unit size would have been chosen as a reference. With the low pressure sodium system, single reactors of two or three times the output are certainly feasible, and should result in further improvements in generation costs.

Steam Conditions and Boiler Design

It seems fairly clear, so far as fossil fuelled plants are concerned, that unless there are some unforeseen developments in the next 6 or 7 years in the cost or high temperature properties of materials, stations commissioned in the late 1970's will have steam temperatures no higher than 560°–570°C and with the lower incentive for capital investment in high temperature plant resulting from the recent break-through in nuclear power, top temperatures may not, in fact, exceed 540°C.

If this is the case for high fuel cost plants, then it seems unlikely that the low fuel cost fast breeder will optimize at a higher temperature, and it might well optimize at a lower, although it would be prudent at this stage to assume 560–570°C as desirable. Steam pressures are most likely to be dictated by the steam side economics alone, since the good heat transfer properties of sodium, coupled with the low pumping power, mean that a high core inlet temperature can readily be achieved.

In the reference double sodium circuit design, the steam temperature of 560–570°C is achieved with a core mixed mean outlet temperature of 610°C, although with a somewhat more generous design of primary heat exchanger this would be reduced to 600°C.

A secondary sodium circuit has been included because it is not yet clear that an adequately safe and operationally acceptable design, involving direct heat transfer from the primary sodium to the steam side, can be proposed. Extensive work on the frequency and modes of failure of various boiler designs, and on the detection and control of subsequent reaction products, is necessary before such a step can be taken, though such a design might well show attractive savings in plant cost coupled with a reduction in sodium tmperature.

The adoption of a 610°C mixed-mean outlet implies rather better mixing in the fuel subassemblies than has been assumed for the prototype, and might also necessitate the use of continuously variable gagging in the core as well as the breeder. However, it is not unreasonable to expect some relaxation in cladding failure criteria as development continues and experience is accumulated. Alternative fuel concepts, involving venting of the fission gases, might permit considerably higher hot spot temperatures. In any case, if 610°C were the optimum, the penalty for dropping 20°C or so (if this proved necessary) would not be very severe; a recent analysis indicates some 0·017 mills/kWh penalty for a 2×1000 MW(e) station.

The steam generator is, for obvious reasons, one of the key items in the development of sodium cooled reactor systems. The high sodium heat transfer coefficients leading to heat fluxes of the order of 130 W/cm^2 offer attractive reductions in surface area compared with the gas/steam system, but the nature and possible consequences of the sodium–water reactions imply that manufacture and inspection techniques will need to be of much higher standards than for conventional boilers, to minimize leakage. In the past this has led to double-wall or double-tube design, with liquid or solid bonds between, but such designs tend to be expensive as well as to increase the difficulties of achieving high temperatures. The single-wall tube is much to be preferred and, although some leakage of water into sodium is inevitable, it is hoped that careful control during manufacture will keep the incidence of leakage down to operationally and economically acceptable values. The frequency of failure and rates of leakage that can be so designated are at present unknown. Leaks direct into sodium, where the reaction products have direct access to the flaw, are considered particularly undesirable, and for this reason present designs with U-tubes and annular headers aim to keep all tube-to-tube and tube-to-tube-plate welds in the cover gas space. Careful control of impurities will be necessary on both steam and sodium sides to limit inherent corrosion; full flow demineralization is proposed for the feed-water, and cleanup loops are incorporated in each secondary sodium circuit.

Ferritic material is necessary in evaporation regions and uncertainty on the sodium side in the stability of ferritics (with respect to carbon loss at intermediate temperatures experienced at part load in the once-through sub-critical system, and consequent carburization of austenitics in the higher temperature regions) was one of the factors contributing to the original selection of a Lamont type boiler, with the evaporator, superheater and reheater all arranged in separate units. Considerable confidence has developed in this design, which permits a wide choice of materials for the different sections without the necessity of transition welding. More recently, information has been accumulating on the potentially satisfactory behaviour of 9% and 12% chrome steels for use in all sections of the steam generators, and more attention is being given to the possibilities of the once-through system, both sub- and supercritical.

Breeding Gain Fissile Rating and Fuel Cycles

With regard to breeding gain and fissile rating, which together determine fuel doubling time, it is not possible this far ahead to forecast where the optimum figures will lie, since there are uncertainties in installation programmes, in breeding prediction, and in the effects on costs and performance of measures aimed at improving breeding performance. However, certain general points can be made.

First, the fast breeders will be built only because they are competitive and, as such, there will be incentive to use them for as much of any new installation programme as possible; second, in the early years plutonium will be in short supply and even if the uranium-235 fuelled fast breeder turns out to be the nearest competitor, it will still be significantly more expensive. These points imply that target ratings should be high, and target fuel doubling times significantly less than those estimated for power demands.

In the major electricity consumption areas of the world, consumption has been increasing at rates of between 4·5% and 10% per annum in recent years.[3] In the United Kingdom the present rate is about 6½% per annum, corresponding to a system doubling time of a little over 11 years. Whilst electricity growth rates of these orders may persist for the next decade, it should be noted that the total energy requirements in the same areas are expanding at a much slower rate, so that some flattening off in the electricity demand curves may ultimately be expected.

For the reference design of a commercial unit, with a 91·5 cm active core length and 23 cm of breeder at either end (i.e. similar to that proposed for the prototype in 1965) and two rows of radial breeder, the breeding performance has been estimated to give an exponential fuel doubling time of about 10 years with a continuous refuelling cycle, although there is considerable uncertainty in this figure, given the data and methods of calculation available at present. From the previous paragraphs, a 10 year doubling time would appear somewhat long for an early commercial unit. Further calculations have indicated that with slightly shorter fuel length and twice as much axial breeder, the exponential doubling time could be reduced to below 6 years. The pin length would increase a little, but with careful design and a small penalty in core pressure drop, this could probably be incorporated within much the same overall subassembly length.

Further improvements in breeding gain may arise from the eventual successful development of carbide fuel, and perhaps from the use of pure metal for much of the breeder. Recent work within the UKAEA[4] has indicated that for large cores of up to 6000 litres, improvements in breeding gain of 0·15 may be obtained with the former, and a further 0·1 with the latter. Taken together, these improvements, if realized, might lead to exponential doubling times of below 4 years. However, the uncertainties in present calculations should not be overlooked.

There is also scope for a more comprehensive study of fuel and breeder cycles and of the use of reflectors, between core and breeder, as proposed in the reference design. Where off-load refuelling has been adopted in that design, a limited survey has shown that the economic compromise between the best utilization of fuel, associated with a continuous cycle, and the lowest outage cost, associated with single batch discharge, may lie in a 75-day cycle, with the inner and outer enrichment zones being discharged in five and six batches, respectively. Similar work in relation to the radial breeder indicates that for the inner row a dwell time of 4 years, with one intermediate rotation, may be the optimum. For the outer row, a 10-year dwell time is indicated, and, because of the high fabrication cost for a low return associated with the long time to approach equilibrium, a single batch discharge is proposed.

All these surveys depend upon some assessment of the worth of breeding. In this connexion, it is perhaps worth noting that since with the sodium system the core power density is high, and the coolant pumping power is small, the capital costs of the station will move relatively slowly with changes in breeding performance, unless, perhaps, at some point a step change in safety requirements were to arise. As a result the effect of breeding will be seen mainly in the changes in fuel cost, itself dependent on any assumptions made regarding the value of plutonium. Since the fuel costs are themselves small, the generation cost characteristic will be comparatively flat, but with the optimum point sensitive to the assumed plutonium value. For the latter reason, surveys on breeding should not be based on arbitrary values for plutonium. A better method is undoubtedly that used by Iliffe,[(5)] where the survey embraces the complete system, or systems, in which the plutonium consumed is limited to that produced.

In the event, the economic worth of breeding will have to be assessed by surveying the best available data at the time. It is to be expected that the customer will make his own assessment and stipulate specific requirements or ground rules with his inquiry.

Station Availability and Fuel Handling

Since the cost penalty for station outage in a large network arises directly from the difference in fuel costs between the outgoing station and that brought in to take up the load, the incentive to reduce downtime will be larger with a low fuel cost fast breeder than with any other system. With a 2×1000 MW(e) station and with a possible outgoing to incoming fuel cost differential of 3·5 mills/kWh, an annual outage of 5% throughout a 20-year life, discounted back at $7\frac{1}{2}$%, represents some \$29 million at the time of construction; so considerable capital investment in on-load or very rapid off-load refuelling may be worth while. For the same reason, with those components which may be liable to more frequent faults, a substantial amount of diversification may be justified.

In practice, reliable on-load refuelling appears very difficult, if not impossible, to achieve. There are severe problems in adequately cooling the highly rated subassemblies with low pressure gas, immediately after they have been removed from the core. The close packing of the core makes single channel access very complicated. In addition, upward flow, which, coupled with a low blanket gas pressure and with no "below sodium" penetrations, has important safety attractions, requires that the safety instrumentation be above the core, which, in turn, makes handling on-load in the hot sodium pool a very complicated operation.

Downward discharge of the fuel involves a somewhat more elaborate handling facility, if tank integrity standards are to be maintained, and as such probably implies that emergency top access would still be desirable. Quite apart from the increase in tank size that would result, any fuel support system arranged for two-way discharge would be likely to necessitate some core dilution to allow for support grid ligaments.

It is estimated that the off-load scheme adopted for PFR could be designed, when applied to the commercial reactor, to give an outage penalty of about 2%, most of this occurring during periods of low demand at week-ends, when the cost of replacement power is less. Under these circumstances, the extra complexity and additional safety problems associated with on-load handling are likely to outweigh the apparent reward.

Safety

With the increasing size of the nuclear installation, the fast breeders of the late 1970's, in common with other nuclear plant, will need to be built in areas of higher population density, requiring even higher safety standards than those currently accepted for more remote sites.

In considering containment, the dominant feature of the fast reactor is, perhaps, that the reactivity will increase if the core is compacted. This characteristic has suggested that certain accidents (such as core meltdown) could possibly lead to a minor nuclear explosion, requiring massive containment to withstand the initial shock, coupled with a very difficult subsequent decay heat removal problem. Recent work by Hicks[6] and others, however, indicates that even for core collapse rates considerably in excess of those that might be expected from a gross meltdown, the energy yield will not be large —in the region of a few tens of pounds TNT equivalent for a 1000 MW(th) core—and model tests conducted by UKAEA have shown that the structures required for normal mechanical duties are largely adequate to contain the incident. Obviously, each new design will need to be carefully examined to show beyond all doubt that there is a negligible probability of some lesser accident leading, in turn, to a very large energy release, but these present signs are encouraging and imply, first, that the cost of any additional containment strengthening will not be large, and second, that it should not be unduly difficult to provide a decay heat removal system which will remain intact.

In the event of cladding failure or local fuel meltout the affinity of sodium for iodine will significantly reduce the activity reaching the blanket gas, where the low pressure will again facilitate containment of the remaining gaseous and volatile fission products.

However adequate the containment, the customer will not take kindly to a design which contains any significant probability of an accident leading to the severe damage or write-off of his capital investment. The aim of the designer must be to achieve such a higher degree of integrity in the core, core support structure, primary coolant circuit, impurity control, fault detection, and shutdown equipment, that all faults which could lead to such an occurrence have themselves a negligible probability. In this respect the sodium cooled, single tank system has a number of attractive features. The low pressure and the absence of stress raising penetrations below sodium level make it possible to design a tank and leak jacket with a large safety factor comparatively cheaply, and the large thermal capacity of the primary coolant forms a substantial heat sink (following loss of normal heat rejection circuits) allowing a considerable time lapse before emergency equipment need be brought into operation.

3. Costs

Competition

Fast breeders will only be built if they are economically competitive. It seems reasonably clear that the competitors are likely to be gas cooled, or possibly water cooled, thermal reactors in the large sizes, with fossil fuelled plants at the smaller end of the range, extending, in areas where such fuel is cheap, to the large sizes as well. It is not easy to forecast what the costs associated with these systems will be in the late 1970's and it is safest to use the figures that exist today and to note that some margin will have to be allowed for future developments.

Quoted cost figures are always liable to misinterpretation because of the way in which costs—particularly capital costs—for apparently equivalent plant appear to vary from country to country. However, as long as all figures relate to plant designed and costed to the same known ground rules in the same country, the comparison should be generally valid and readily amenable to any adjustment for other conditions in other countries. Accordingly, in Fig. 1 the generation cost for Dungeness "B" 5·4 mills/kWh has been plotted and extrapolated for sizes up to 2×1000 MW(e), using a two-thirds power law for the capital costs in the absence of firm figures for large units, whilst holding fuel costs constant, and using the same ground rules as in the Dungeness "B" assessment.[7] Also plotted is a band for coal and oil fired stations, with some recently quoted figures[8] identified.

So far as the applicability of these curves to the late 1970's is concerned, some reduction—perhaps as much as 10%—in the thermal reactor figures must be expected, as design and performance improve, and from recent assessments[9] it would seem unwise to assume that this will be in any way offset by rising fuel costs associated with scarcity of cheap uranium ore. In the coal and oil fired stations, however, where the fuel costs are dominant, it seems at present unlikely that any significant reductions will arise.

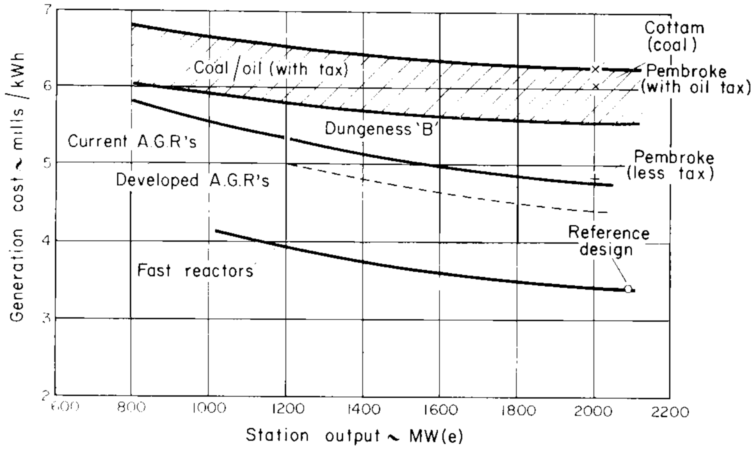

FIG. 1. Comparative costs.

Costs for the 2 × 1000 MW(e) Reference Design

The reference design described briefly in section 4 of this paper has been worked up in considerable detail and costed. In preparing the costs, it has been assumed that all major development has been completed as part of the prototype reactor programme and, therefore, only charges for normal manufacturing development are included. The capital costs quoted have been determined on the basis of the construction of one 2 × 1046 MW(e) station only, and no unit cost reductions due to increased volume of production have been assumed. The fuel costs relate to a 12,000 MW(e) programme of fast breeder installation. In evaluating the generation cost, the recommendations for the Dungeness "B" station tender have generally been followed. The major points to note are that the station life is assumed to be 20 years, the interest rate 7½%, and the load factor 75% for a station with on-load refuelling. For off-load refuelling a debit is made equivalent to the cost of electricity produced elsewhere during the additional outage involved. In keeping with the remarks under the heading Breeding Gain Fissile Rating and Fuel Cycles, zero value has been assumed for plutonium, although with the particular inventory, breeding gain and interest rate involved here, the effect on generation cost of varying the value up to $28/g is only some 10%.

It is emphasized again that the costs relate to U.K. conditions and should be compared with U.K. prices for competitive systems.

The cost figures are:

Capital costs		$/kW
Primary circuit (including primary heat exchangers)		10·8
Secondary circuit and boilers		25·4
Generating plant, steam and feed piping and CW system excluding customer's supply		38·4
Buildings and civil works		28·7
Auxiliary systems associated with sodium		4·2
Other auxiliaries and main transformers and switchgear		14·7
Fuel handling		3·0
Control and instrumentation		7·1
Therefore, the tender price (This figure includes on-costs covering design and project management services, financing charges, insurance, contingencies, etc.)	=	132
Interest during construction (assume approximately 10% of tender price)		13
Therefore, tender construction cost	=	145
Customer's direct costs and fees during construction (including cost of land, site development, consultant's fees, etc., and CW off-shore works, pumphouse, and syphon recovery chamber)		9·4 (typical)
Therefore, total construction cost	=	154·4
Adjustment for load factor (present worth basis)		5·6
Customer's other works costs (present worth basis)		14 (typical)
Fuel costs (present worth basis)		57
Total	=	231
Therefore, tender cost of generation	=	22·4 $/kW per annum
	=	3·4 mills/kWh

Before discussing the implications of these costs, a word or two about their accuracy and make-up. In an attempt to achieve a feel for the uncertainty, a number of comparative estimates have been obtained for as many of the major novel features as possible, and an assessment of these indicates that the overall accuracy range is some $\pm 10\%$. Where discrepancies arose, the figures selected have in general erred on the generous side, so that the costs quoted are somewhat above the indicated mean.

When preparing the detailed design for costing, the authors were conscious of the fact that the reality frequently turns out to be somewhat more costly than the expectation. With this in mind, rather more elaboration than might strictly be necessary has been built into a number of the specifications prepared for estimating. This policy is reflected, for instance, in the costs for the sophisticated computer control system, and in those for the highly diversified boiler plant.

Referring again to Fig. 1, the generation cost of 3·4 mills/kWh has been plotted and extrapolated for smaller sizes to show a possible trend for comparison with the AGR which, it may be recalled, showed a marginal cost advantage over the BWR, for Dungeness "B". The extrapolation has again been based on a two-thirds power law for capital costs and assumed constant fuel costs. Since the core physics does not vary substantially in going from PFR to the 1000 MW(e) core, fuel costs would not be greatly changed in this range; the slightly higher enrichment for the smaller sizes being perhaps offset by a somewhat harder spectrum and by improved internal breeding.

If the trend is correct, the sodium cooled fast breeder should be competitive over a very wide range, with a margin in excess of 1·2 mills/kWh over the current AGR's, and perhaps 1 mill/kWh over developed AGR's.[7]

Of the difference, in the range 1000–2000 MW, some 40% is due to reduction in capital cost. In is difficult to compare details of two systems so radically different, but certain pointers can be identified. The high heat transfer properties of sodium lead to a big reduction in boiler surface, which more than offsets the extended use of stainless steels. Similarly, the low sodium pumping power results in comparatively low cost pumps and drives. Finally, the low sodium system pressure does much to simplify the design, and although stainless steels have been extensively—and perhaps sometimes unnecessarily—used, the sections are usually thin and construction relatively easy.

Development Potential

The reference design was by no means fully optimized and some improvement in costs can certainly be expected from a more careful study of the importance of the major parameters. Quite apart from this, however, improvement may be expected in many areas, including the use of low-alloy steels in the lower temperature regions of the sodium circuits, and, with increased confidence in component performance, a less accessible but more compact primary circuit layout. Proven component performance will also lead to less diversification of primary heat exchangers, pumps, and boiler units, to the possible use of once-through boilers, and perhaps, in the longer term, to the elimination of the secondary sodium circuit.

Neglecting the latter, the potential savings in capital cost have been estimated as at least $16/kW in a 2 × 1000 MW(e) design. On the fuel side future developments, such as the use of carbides and vented fuel elements, could well lead to reductions of 0·25–0·35 mills/kWh so that generation costs of some 2·8 mills/kWh may ultimately be possible for this size of plant.

Hitherto it has usually been assumed that the initial shortage of plutonium will limit the rate of installation of fast reactors and that the deficit will have to be made up with thermal reactors. If the difference between these two

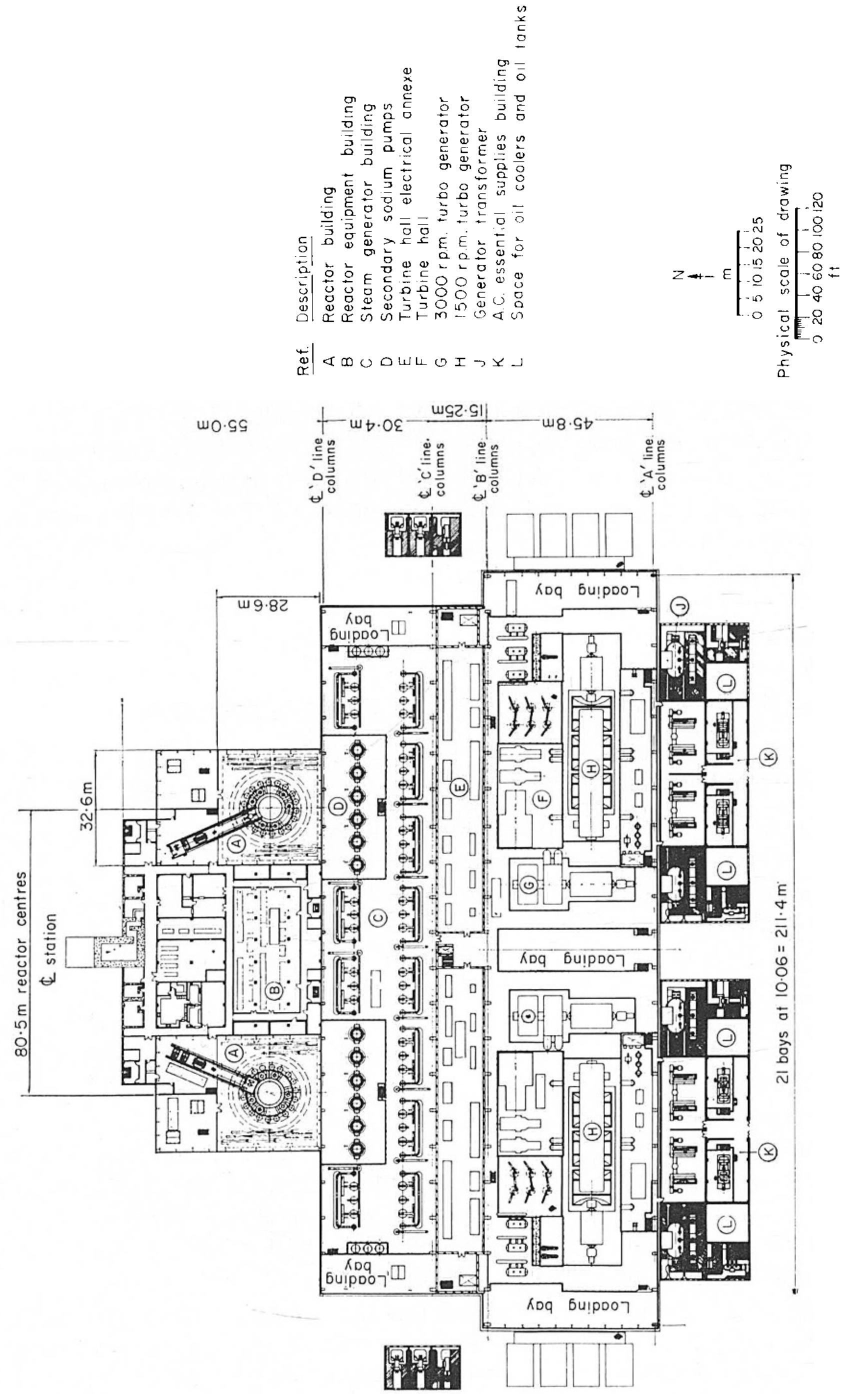

FIG. 2. General arrangement of station at operating floor level.

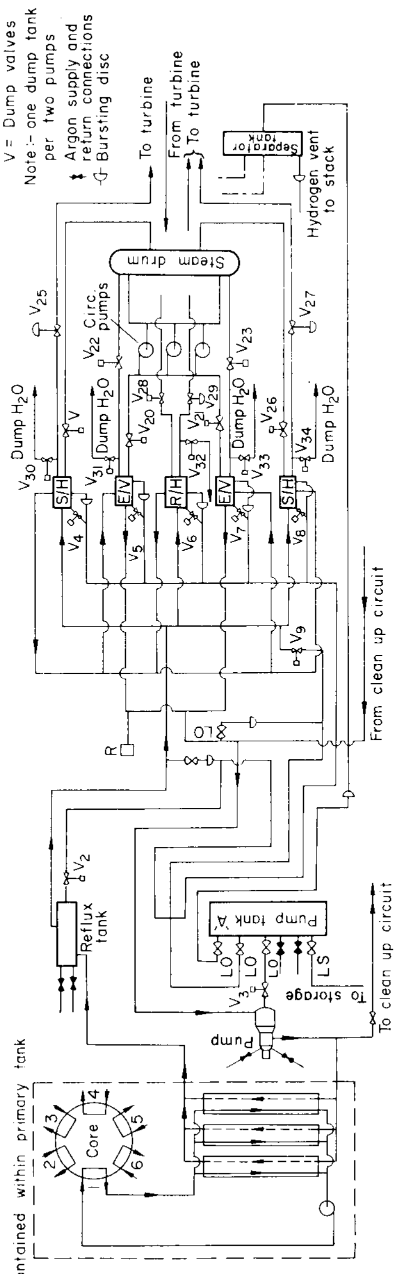

FIG. 3. System flow diagram.

types turns out to be as large as may be inferred from the figures discussed in this section, it seems possible that a more economic solution will be to use U^{235} fuelled fast breeders instead of the thermal reactors.

4. Outline Description of the Reference Design

General

The reference design is for a 2×1000 MW(e) station. As shown in Fig. 2, the reactors, steam-generating plant, turbo-generators, and auxiliaries are all arranged to be in one building. The reactors are contained in separate halls, linked by a common reactor services block.

The reactor thermal output is transferred to the primary sodium circuit,

TABLE 1. REFERENCE DESIGN PARAMETERS

Net electrical output per reactor		1046 MW
Overall plant efficiency		44%
Core thermal power output	at equilibrium	2272 MW
Breeder and reflector thermal power output	at equilibrium	107 MW
Total thermal power per reactor		2379 MW
Core and breeder size and performance		
Height of active core		91·5 cm
Axial breeder length at each end		23 cm
Diameter of active core		269 cm
Diameter over radial breeder		350 cm
Core L/D		0·34
Core volume		5210 litres
Number of subassemblies:		
Core fuel		195
Radial breeder		150
Control		18
Reflector		66
Initial fuel loading (PuO_2+UO_2)		15·1 tonnes
Initial Pu loading (metal)		2·4 tonnes
Feed enrichment % PuO_2, for 75 days, fuel cycle:		
Zone 1		15·4%±1%
Zone 2		20·8%±1%
Maximum rating of fuel		234 w/g oxide
Mean rating of fuel		150·4 w/g oxide
Average core power density		436 kW/litre
Radial averaging factor		1·29
Axial averaging factor		1·2
Overall averaging factor		1·55
Exponential fuel doubling time		10 years
Fuel and axial breeder subassembly		
Outside dimension, across flats of hexagon		14·1 cm
Pitch		14·5 cm
Number of fuel pins		325
Pellet o.d.	or 8·8 g/c³ vibro fuel	5·08 mm
Pellet i.d.	or 8·8 g/c³ vibro fuel	1·52 mm

TABLE 1 (*continued*)

Active fuel length	91·5 cm
Void length	121 cm
Pin pitch (triangular)	7·37 mm
Peak can temperature (mid-wall)	670°C
Design max. burnup	10%
Weight of fuel in subassembly	51·2 kg
Weight of axial breeder per subassembly	28·13 kg
Radial breeder subassembly	
Outside dimension	14·1 cm
Pitch	14·5 cm
Number of pins	91
Radial breeder material	UO_2
Pellet o.d.	1·24 cm
Pin o.d.	1·35 cm
Active length	137 cm
Void length	118 cm
Pin pitch (triangular)	1·39 cm
Weight of UO_2 in subassembly	140 kg
Control (tantalum rods)	Reactivity investment
Power—Doppler	0·7%
Power—expansion of fuel	0·1%
Fuel burnup (10% max.)	1·6%
Sub-criticality margin	1·0%
Withdrawn/stuck rod	0·5%
Basic requirements	3·9%
Additional requirements for uncertainties in calculation	1·5%
Primary sodium	
Outlet temperature from core	610°C
Inlet temperature to core	380°C
Flow rate in core	29×10^6 kg/h
Flow rate in breeder	$0{\cdot}52 \times 10^6$ kg/h
Primary circuit pressure drop	4·2 kgf/cm^2
Primary circuit pumping power	6·4 MW
Secondary sodium	
Inlet temperature to primary heat exchanger	340°C
Outlet temperature from primary heat exchanger	580°C
Flow rate	$28{\cdot}4 \times 10^6$ kg/h
Secondary circuit pressure drop	7 kgf/cm^2
Secondary circuit pumping power	9 MW
Steam side data	
TSV pressure	163 kgf/cm^2
TSV temperature	560°C
Number of reheat stages	1
Reheated steam inlet pressure	39 kgf/cm^2
Reheated steam inlet temperature	560°C
Superheated steam flow	$3{\cdot}16 \times 10^6$
Reheat steam flow	$2{\cdot}48 \times 10^6$ kg/h
Feed temperature	252°C

through heat-exchangers to a secondary sodium circuit, and thence to the steam-generators, producing steam conditions at the turbine stop-valve of 163 kgf/cm^2 and 560°C. A flow diagram for the system is shown on Fig. 3 and a list of the major parameters appears in Table 1.

Reactor and Primary Circuit

The entire primary circuit, shown on Fig. 4, is immersed in sodium in a stainless steel tank some 17·4 m in diameter by 14·3 m deep. All components and the tank itself are suspended from the vault roof.

The core, core reflector, and breeder assembly, forming a hexagonal prism some 3·66 m across flats by 3·81 m high, is made up of 511 hexagonal subassemblies each 14·1 cm across flats on a 14·5 cm pitch. A plan of the arrangement is shown on Fig. 5.

Each core zone subassembly consists of 325 2·74 m long stainless steel clad pins, of 5·84 mm o.d. and 0·381 mm wall thickness, containing 91·5 cm of mixed plutonium uranium oxide, with 23 cm of natural or depleted uranium oxide breeder at each end, and with a 122 cm void for fission gases at the lower end. The pins, supported by grids at frequent intervals, are encased in a hexagonal, stainless steel wrapper. Radial breeder subassemblies are of similar construction but with fewer, fatter pins. The subassemblies are supported at the lower end by spikes fitting into a stainless steel support grid.

Activation shielding surrounds the core, and the entire assembly, mounted on the support grid, is hung from the vault on a support skirt.

Sodium at 380°C is pumped by six single-stage centrifugal pumps from the sodium pool to the core support grid, which acts as a pressure plenum, and thence through the subassemblies, emerging at 610°C. The hot sodium flows radially between the hangers of the core support skirt and into the hot launder, where it is distributed to the eighteen primary heat-exchangers.

These are of the straight-through, shell-and-tube type, and the primary sodium flows through the shell side, under the action of the 2·14 m head difference between the hot launder and the cold sodium pool.

A feature of the primary circuit is the guarantee of integrity against loss of sodium. To achieve this, the stainless steel tank has a mild steel jacket, and the interspace between is monitored for leaks in either. Both tank and jacket are designed to have low stresses, and there are no penetrations or stress raisers below the sodium level.

Secondary Sodium Circuit, Steam Generator, and Turbo-alternator

Each reactor has six independent, secondary sodium circuits, each feeding a recirculation steam generator. The output from the six steam generators feeds one cross-compound, turbo-alternator set. For each of the sodium

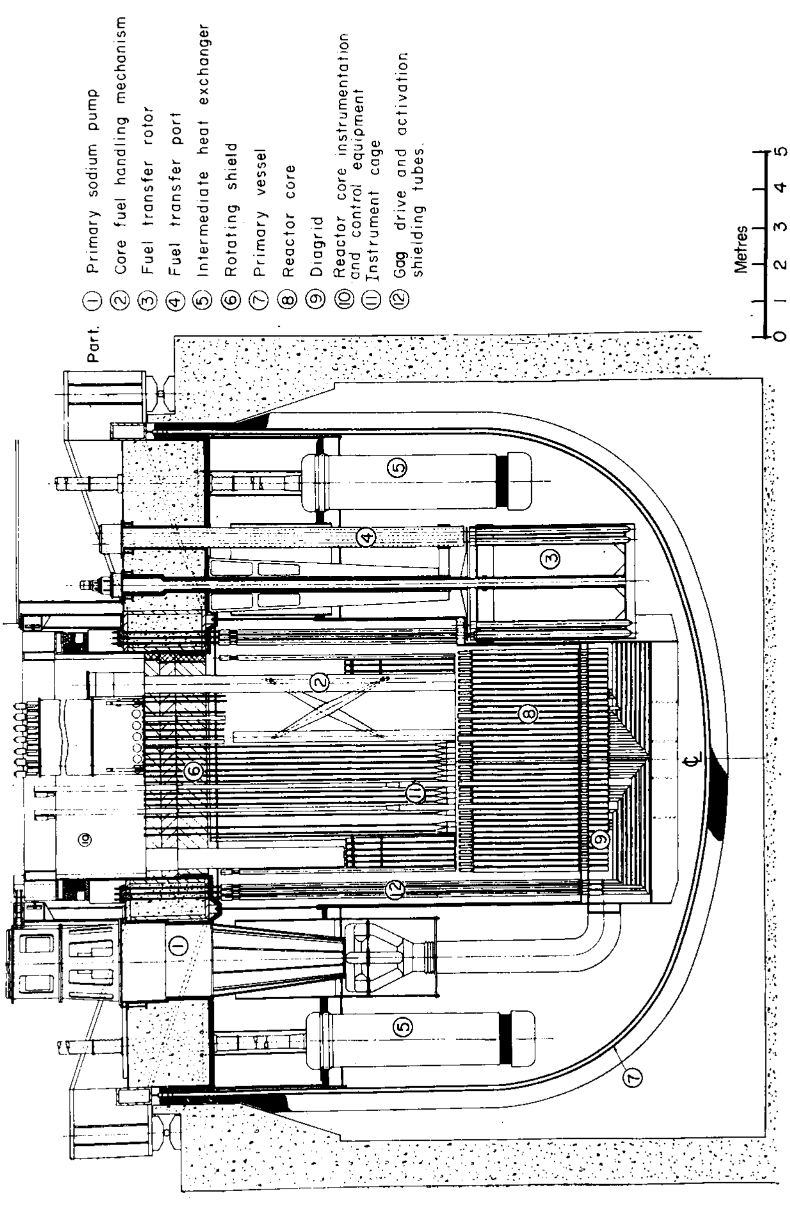

FIG. 4. Sectional elevation through reactor circuit.

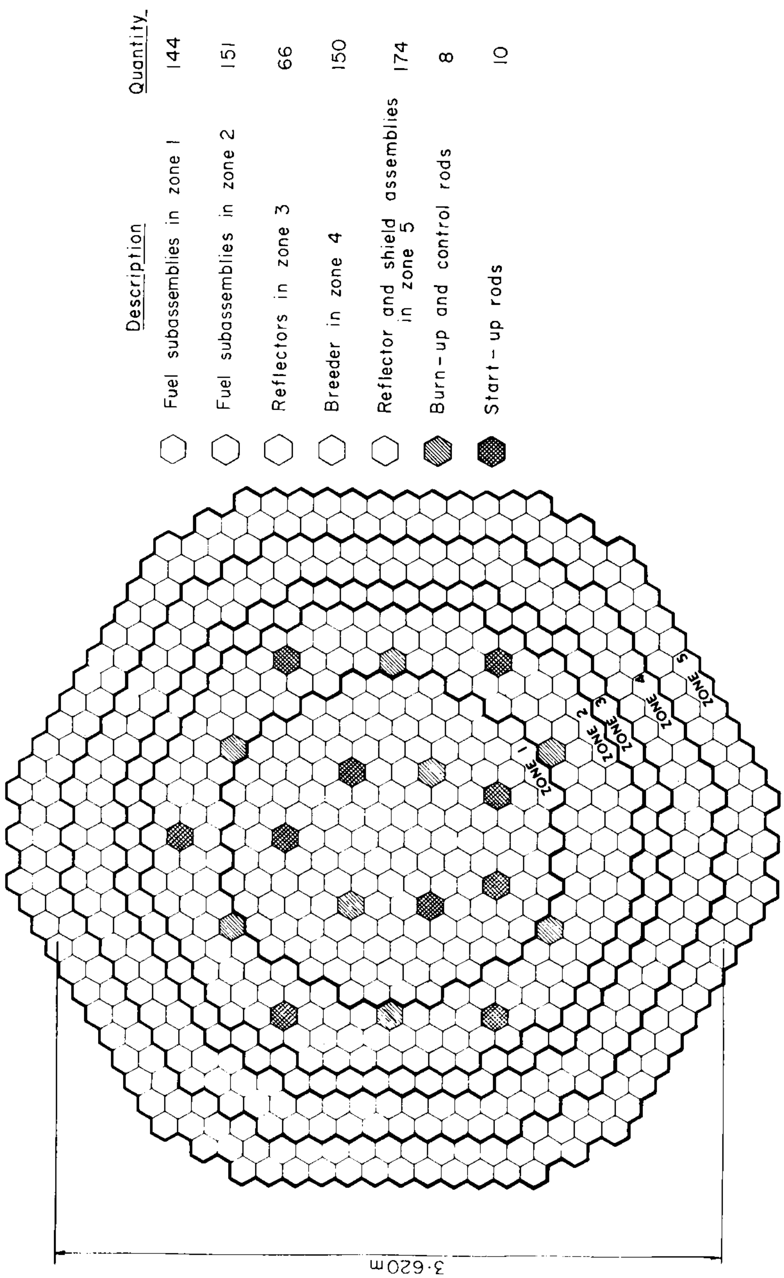

FIG. 5. Core utilization diagram.

circuits, pipe connexions from three primary heat-exchangers pass through plugs in the vault roof, and are then bussed into common return and flow pipes linking the reactor with the boiler house. The hot sodium (at 580°C) is passed through a reflux tank and then to a five-unit steam generator, whence it flows through one reheater and two superheater units all in parallel, followed by two evaporator-economizer units in parallel. Outlet sodium (at 340°C) from these units is piped to a secondary sodium pump (of similar design to the primary pumps), and thence to the reactor.

All sections of the steam generator are of similar shell-and-U-tube construction with annular heads, with sodium on the shell side. Austenitic, stainless steels are used throughout the superheater and reheater units, whilst the evaporators have 9% chrome/molybdenum tubes, ferritic steel headers, and low alloy, stainless clad shells.

Feedwater at 179 kgf/cm^2 and 252°C is passed to the economizer section of each evaporator, to be heated to 312°C and then to enter the downcomer header, upstream of the boiler circulating pumps. The resulting slightly sub-cooled water is circulated through the two evaporator sections of the steam drum. Dry, saturated steam from the drum is passed through the superheaters, leaving at 173 kgf/cm^2 and 566°C.

The superheated steam from the six steam generators feeds the HP cylinder of one cross-compound, turbo-alternator unit. Expanded steam from the HP cylinder is reheated to 566°C at 40·3 kgf/cm^2 and then fed to the IP and LP cylinders. A portion of the HP exhaust steam is bled off for auxiliaries and feed heating.

Because occasional tube failure is inevitable and results in a sodium–water reaction, the steam generator units are fitted with bursting disks, venting to a dump system. In addition, quick-acting dump valves are fitted on the sodium side, and isolation valves on the steam and water side.

Sodium Circuit Auxiliaries

The free sodium surface in the primary tank is protected by an inert atmosphere of argon. The gas is at 1·07 kgf/cm^2 to prevent any significant in-leakage of air, and all plugs and components passing through the vault roof are fitted with a double seal.

The secondary sodium circuit and sodium supply, dump and storage circuits, are protected in a similar manner.

To minimize the amount of oxygen and moisture reaching the sodium, dryers and scrubbers are provided with the argon system.

Sodium cleanup loops are provided for the primary and secondary sodium circuits, and for the sodium storage system. Each primary circuit has three loops, each of 1 m^3 per minute capacity, consisting of a pump, two regenerative heat-exchangers, and cold traps designed to keep the oxide level down to

20 ppm. Similar loops are attached to each secondary circuit, but there the main pump head is used for circulation.

Electrical trace heating is supplied to certain sodium pipework and storage tanks. The primary tanks are heated, during filling, by fitting temporary heaters to the argon blanket system; and the steam generators by using auxiliary steam

Items of plant contaminated with sodium, and needing cleaning, are transported in argon-filled flasks to cells for steam cleaning, washing-down, and drying. One of the cells has an organic fluid bath for cleaning delicate items.

Fuel Handling

The reactor is shut down for some 40 hours during a week-end once every 9 to 10 weeks for refuelling. During the shutdown about fifty irradiated fuel subassemblies are transferred, under sodium, from the core to an in-vessel decay store known as the transfer rotor, and replaced with new fuel previously stored in the rotor. The transfer is effected using the core fuel-handling mechanism mounted on the periphery of the rotating shield, which is itself mounted above the core in the pilecap.

Irradiated fuel is left for some 30 days in the transfer rotor then removed (with the reactor on load) by the charge machine via the access ports in the vault roof, and, undergoing forced gas cooling, is transported to a decontamination cell where it is steam cleaned and carried underwater to either the dispatch bay or the storage pond.

After inspection, new fuel is moved to an argon purge cell from which it is collected by the charge machine and loaded into the transfer rotor, ready for the next shutdown.

Control and Operation

Each reactor has eighteen tantalum absorber rods distributed over the core, meeting a total reactivity requirement including uncertainties, of 5·4% $\Delta k/k$. Of the eighteen rods, ten are used to bring the reactor from subcritical up to power, six are used as burnup rods, and the remaining two are automatic control rods actuated in response to changes in reactor outlet temperature.

A block schematic of the control scheme is shown in Fig. 6. Control of the station has been arranged to allow operation at constant reactor power output, or in a frequency compensating role. The thermal capacities of the boiler and sodium circuits are used to shield the reactor from short term transients arising from small frequency fluctuations.

Reactor inlet temperature is held constant and reactor power changes are initiated at the primary sodium flow controller, with a programmed adjustment of the set point on the outlet temperature controller governing the

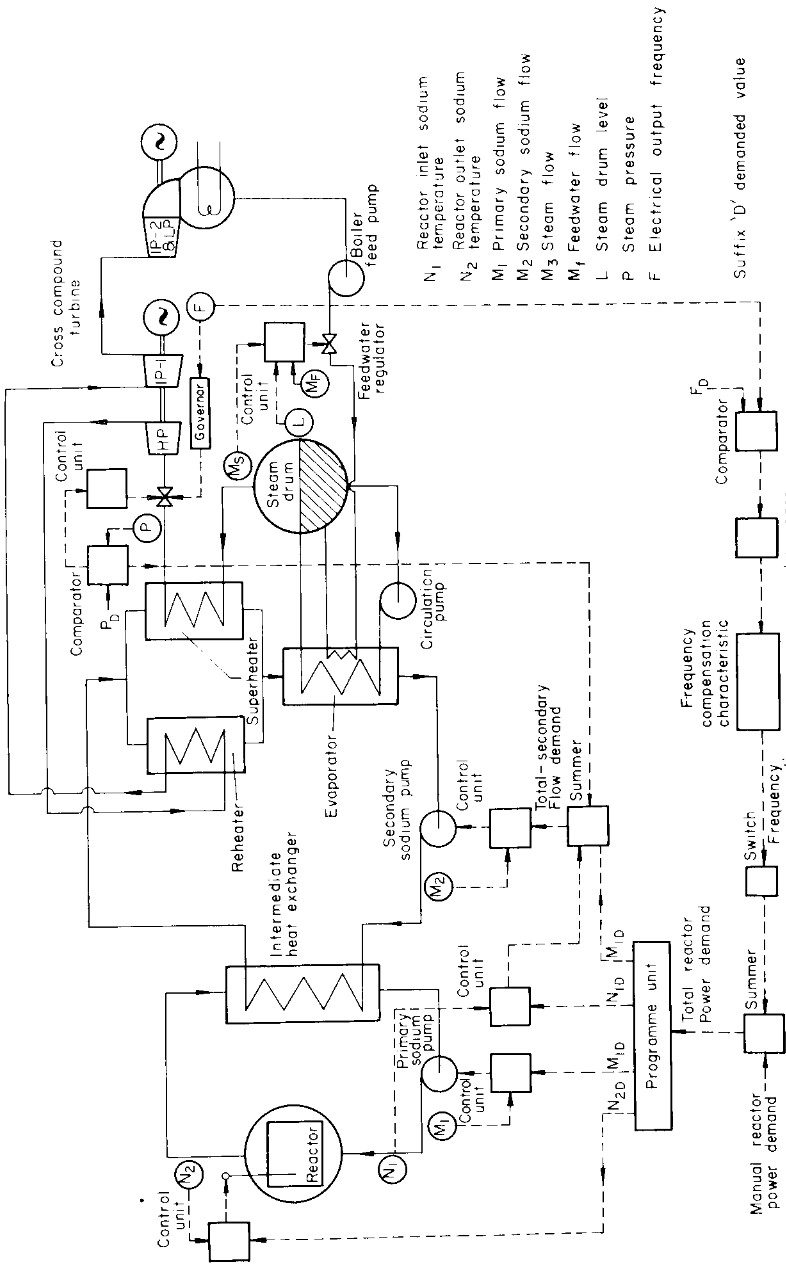

FIG. 6. Control diagram.

position of the two automatic control rods, in order to maintain constant steam temperature. The secondary sodium flow is controlled by a three-term controller; the first component being a demand signal directly proportional to primary flow, supplied by the reactor power programme unit. The second component is a small trimming term which slowly overrides the first to maintain reactor inlet temperature constant, and the third component is proportional to the difference between the steam pressure and the demanded value, and gives a transient increase in secondary flow if steam pressure falls. Primary and secondary flows are adjustable down to 20%

Steam pressure is controlled by turbine throttle speeder gear movements and the water level in the drum is maintained by adjustments of the feedwater regular valve using a three-term control law which also takes into account the water and steam flows in and out of the drum, respectively.

References

1. PROF. DR. DR. H. MANDEL, The use of nuclear power stations in the electricity grid, *2nd Foratom Congress* (Sept. 1965).
2. G. F. KENNEDY, United States power plant design trends: 1965, *Proc. Inst. Electrical Engineers*, **113**, No. 1 (Jan. 1966).
3. United Nations Statistical Papers, Series J.
4. T. P. MOORHEAD and A. S. B. BELCHER, *Dependence of Breeding Upon Some Design Parameters of Large Fast Reactors*, UKAEA Report AEEW—M534.
5. C. E. ILIFFE, *Economics of Plutonium Breeding—Plutonium Inventory and Uranium Conversion*, UKAEA Report TRG1207 (R).
6. E. P. HICKS and D. C. MENZIES, Theoretical studies on the fast reactor maximum accident, *Proceedings of the International Conference on Safety, Fuels and Core Design in Large Fast Power Reactors, Argonne National Laboratory (October 11–14 1965).*
7. *An Appraisal of the Technical and Economic Aspects of Dungeness "B" Nuclear Power Station*, CEGB Publication (July 1965).
8. *Fuel Policy*, Cmnd. 2798, Her Majesty's Stationery Office (Oct. 1965).
9. *World Uranium and Thorium Resources*, OECD—ENGA Report (Aug. 1965).